

BACKLOG SUBJECT EXAMINATIONS – AUGUST / SEPTEMBER 2023

B.E. – Artificial Intelligence and Data

Program	: Science / Artificial Intelligence and Machine Learning	Semester	: III
Course Name	: Data Structures	Max. Marks	: 100
Course Code	: AD33 / AI33	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Implement using C to find the difference between the two complex numbers. CO1 (08)
 - Explain an efficient way of storing a sparse matrix in memory. Write a module to find the transpose of a sparse matrix stored in this way. CO1 (08)
 - Differentiate between Structure and Union. CO1 (04)
- Explain the differences between malloc() and calloc() functions used for memory allocation. CO1 (04)
 - Write a C function to concatenate two strings using pointers which has to be named as strcat_user(), that accepts two string pointer variables. Do not use any string related predefined functions in your code. Your user defined C function should Concatenate two strings, then return the pointer to the destination string. Use only character pointers for accessing each character from/to string variables. CO1 (08)
 - Illustrate how a two dimensional array is allocated dynamically, with a suitable example. CO1 (08)

UNIT - II

- Write a C function to reverse a string using stack. CO2 (06)
 - Write a C program to evaluate the following postfix expression
62/3-42*+
Tabulate the stack contents. CO2 (08)
 - Describe limitations of a linear queue with suitable examples. CO2 (06)
- Write a C function to perform insert and delete operations on multiple stack. CO2 (06)
 - Why we need to learn postfix and prefix expressions? Where they are used? Give justification for your answer. CO2 (06)
 - Write C functions to perform add and delete operations on a circular queue using dynamic implementation. CO2 (08)

UNIT - III

- Illustrate a doubly linked circular list with header node with a neat diagram. CO3 (06)
 - Write a C function to insert a node after the node X in a singly linked list. CO3 (06)
 - Write a C function that searches for an integer NUM in a circular linked list. If NUM is in the list, function should return a pointer to that node. Otherwise it should return NULL. CO3 (08)

6. a) Justify the merits of linked stack over array implementation of a stack. CO3 (06)
b) Write a C function to invert a singly linked list. CO3 (06)
c) Write a C function to insert a node into a doubly linked circular list with header node. Also explain with suitable example. CO3 (08)

UNIT- IV

7. a) Define Trees. Give the implementation of Trees in C. list the applications of Tree data structures. CO4 (10)
b) Construct a binary search tree with elements 50, 70, 60, 20, 90, 10, 40, 100. Perform the In order, Preorder and Post order traversals on constructed tree. CO4 (10)
8. a) Write a C Function to perform: CO4 (10)
i) Iterative inorder traversal using stack
ii) Level – order traversal using Queue.
b) Construct Max Heap and Min Heap for the Key values CO4 (10)
14, 10, 15, 20, 2, 21.

UNIT - V

9. a) Define Graph. Explain how we can represent Graph as an Abstract Data type with code? CO5 (10)
b) Write C function to perform: CO5 (10)
i) Breadth First Search and
ii) Depth First Search of a Graph.
10. a) Define AVL trees. Give the rules and syntax of single and double rotations in AVL trees. CO5 (10)
b) List the properties of Leftist trees. Construct a leftist heap for 10 elements. CO5 (10)
